Basis Theorem:-

Let, V be a vector space

$S = \{v_1, v_2, --- v_n\}$

if:

→ S should be Linear Independent

→ $Span(S) = V$

If $S = \{v_1, v_2, \cdots v_n\}$ is a basis for a vector space V, then every vector v in V can be expressed in:

$$v = \alpha_1 v_1 + \alpha_2 v_2 + \cdots + \alpha_n v_n$$

Example:-

(i) Show that $v_1 = (1,0,0)$, $v_2 = (0,1,0)$, $v_3 = (0,0,1)$ then $S = \{v_1, v_2, v_3\}$ is basis of R^3.

Sol:-

We've to show that S is Linearly Independent:

$$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = 0$$

$$\alpha_1(1,0,0) + \alpha_2 \cdot (0,1,0) + \alpha_3(0,0,1) = (0,0,0)$$

$$(\alpha_1, 0, 0) + (0, \alpha_2, 0) + (0, 0, \alpha_3) = (0,0,0)$$

$$(\alpha_1, \alpha_2, \alpha_3) = (0,0,0)$$

$$\alpha_1 = 0, \ \alpha = 0, \ \alpha_3 = 0$$

Hence we can say that S is L.I

Condition II:

Now we've to show that Span (S) = V

$$\text{Span}(S) = \alpha_1 V_1 + \alpha_2 V_2 + \alpha_3 V_3$$
$$= (\alpha_1, \alpha_2, \alpha_3)$$
$$= V \in R^3$$

$\therefore \text{Span}(S) = V$

$\therefore S = \{V_1, V_2, V_3\}$ is a basis of R^3.

(ii) Let $V_1 = (1, 2, 1)$, $V_2 = (2, 9, 0)$, $V_3 = (3, 3, 4)$
Show that the set $S = \{V_1, V_2, V_3\}$ is a basis of R^3.

Sol.-

Condition I: For L.I

$$\alpha_1 V_1 + \alpha_2 V_2 + \alpha_3 V_3 = 0$$

$$\alpha_1 (1, 2, 1) + \alpha_2 (2, 9, 0) + \alpha_3 (3, 3, 4) = (0, 0, 0)$$

$$(\alpha_1, 2\alpha_1, \alpha_1) + (2\alpha_2, 9\alpha_2) + (3\alpha_3, 3\alpha_3, 4\alpha_3) = (0, 0, 0)$$

Comparing

$$\alpha_1 + 2\alpha_2 + 3\alpha_3 = 0$$
$$2\alpha_1 + 9\alpha_2 + 3\alpha_3 = 0$$
$$\alpha_1 + 4\alpha_3 = 0$$

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 9 & 3 \\ 1 & 0 & 4 \end{bmatrix}$

$|A| = 1(36-0) - 2(8-3) + 3(0-9)$

$|A| = -1 \neq 0$

The vectors are not zero

So $\alpha_1 = \alpha_2 = \alpha_3 = 0$

So, It is linearly Independent.

Condition II, For Span(S)

Let $(x, y, z) \in R^3$

$\alpha_1 V_1 + \alpha_2 V_2 + \alpha_3 V_3 = (x, y, z)$

$\alpha_1 (1,2,1) + \alpha_2 (2,9,0) + \alpha_3 (3,3,4) = (x, y, z)$

Comparing

$\alpha_1 + 2\alpha_2 + 3\alpha_3 = x$

$2\alpha_1 + 9\alpha_2 + 3\alpha_3 = y$

$\alpha_1 + 4\alpha_3 = z$

Let $A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 9 & 3 \\ 1 & 0 & 4 \end{bmatrix}$.

$|A| = -1$ Solution is consistent

$Span(S) = (x, y, z) \in R^3 = V$

$S = \{V_1, V_2, V_3\}$ is Basis of R^3.

3) Let $V_1 = (1,0,0)$, $V_2 = (2,2,0)$, $V_3 = (3,3,3)$. Show that $S = \{V_1, V_2, V_3\}$ is basis of R^3.

Sol:-

⇒ Condition I: For L.I

$$\alpha_1 V_1 + \alpha_2 V_2 + \alpha_3 V_3 = (0,0,0)$$

$$\alpha_1 (1,0,0) + \alpha_2 (2,2,0) + \alpha_3 (3,3,3) = (0,0,0)$$

Comparing

$$\alpha_1 + 2\alpha_2 + 3\alpha_3 = 0 \quad \text{— (i)}$$

$$2\alpha_2 + 3\alpha_3 = 0 \quad \text{— (ii)}$$

$$3\alpha_3 = 0 \quad \text{— (iii)}$$

From (iii) $\boxed{\alpha_3 = 0}$

Putting α_3 in (ii)

$$2\alpha_2 + 0 = 0$$

$$\boxed{\alpha_2 = 0}$$

Putting α_2, α_3 in (i)

$$\boxed{\alpha_1 = 0}$$

So It is a L.I.

⇒ Condition II: For Span (S)

Let $(x, y, z) \in R^3$

$$\alpha_1 V_1 + \alpha_2 V_2 + \alpha_3 V_3 = (x, y, z)$$

$$\alpha_1 + 2\alpha_2 + 3\alpha_3 = x \quad \text{— (i)}$$

$$2\alpha_2 + 3\alpha_3 = y \quad \text{— (ii)}$$

$$3\alpha_3 = z \quad \text{— (iii)}$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 0 & 2 & 3 \\ 0 & 0 & 3 \end{bmatrix}$$

$|A| = 1(6-0) = 6 \neq 0$

$\because$ Sol is Consistent

Span $(S) = (x, y, z) \in R^3 = V$

$\therefore$ Span$(S) = V$

$S = \{v_1, v_2, v_3\}$ is a basis of R^3

(IV) Let $v_1 = (2, 5, 1)$, $v_2 = (3, 1, 3)$, $v_3 = (4, 0, 7)$

Show that $S = \{v_1, v_2, v_3\}$ is Basis of R^3

Sol:-

$\Rightarrow$ Condition I: For L.I

$\alpha_1 v_1 + \alpha_2 v_2 + \alpha_3 v_3 = (0, 0, 0)$

$\alpha_1 (2, 5, 1) + \alpha_2 (3, 1, 3) + \alpha_3 (4, 0, 7) = (0, 0, 0)$

Comparing

$2\alpha_1 + 3\alpha_2 + 4\alpha_3 = 0$

$5\alpha_1 + \alpha_2 + \overset{0}{\cancel{3\alpha_3}} = 0$

$\alpha_1 + 3\alpha_2 + 7\alpha_3 = 0$

Let $A = \begin{bmatrix} 2 & 3 & 4 \\ 5 & 1 & 0 \\ 1 & 3 & 7 \end{bmatrix}$

$|A| = 2(7-0) - 3(35-0) + 4(15-1)$

$= 14 - 105 + 56 = -35 \neq 0$

Vectors are not zero

So, $\alpha_1 = \alpha_2 = \alpha_3 = 0$

Hence It is L.I

$\Rightarrow$ Condition II: For Span(S)

let $(x, y, z) \in R^3$

$\alpha_1 V_1 + \alpha_2 V_2 + \alpha_3 V_3 = (x, y, z)$

$2\alpha_1 + 3\alpha_2 + 4\alpha_3 = x$

$5\alpha_1 + \alpha_2 = y$

$\alpha_1 + 3\alpha_2 + 7\alpha_3 = z$

$|A| = -35 \neq 0$

Sol is consistent

$Span(S) = (x, y, z) \in R^3$

$Span(S) = V$

$S = \{V_1, V_2, V_3\}$ is basis of R^3

v) Let $V_1 = (1, 2, 3), V_2 = (2, 1, 3), V_3 = (3, 0, 0)$

Show that $S = \{V_1, V_2, V_3\}$ is basis of R^3

Sol:-

$\Rightarrow$ Condition I: For LI

$\alpha_1 V_1 + \alpha_2 V_2 + \alpha_3 V_3 = (0, 0, 0)$

$\alpha_1 (1, 2, 3) + \alpha_2 (2, 1, 3) + \alpha_3 (3, 0, 0) = (0, 0, 0)$

$$\alpha_1 + 2\alpha_2 + 3\alpha_3 = 0$$
$$2\alpha_1 + \alpha_2 \quad = 0$$
$$3\alpha_1 + \alpha_2 \quad = 0$$

$$A = \begin{bmatrix} 1 & 2 & 3 \\ 2 & 1 & 0 \\ 3 & 1 & 0 \end{bmatrix}$$

$$|A| = 1(0-0) - 2(0-0) + 3(2-3)$$
$$|A| = -3 \neq 0$$

Vectors are not zero

So $\alpha_1, \alpha_2, \alpha_3 = 0$

So It is L.I

$\Rightarrow$ Condition II : For Span(S)

Let $(x, y, z) \in R3$

$$\alpha_1 V_1 + \alpha_2 V_2 + \alpha_3 V_3 = (x, y, z)$$
$$\alpha_1 + 2\alpha_2 + 3\alpha_3 = x$$
$$2\alpha_1 + \alpha_2 \quad = y$$
$$3\alpha_1 + \alpha_2 \quad = z$$

$$|A| = -3 \neq 0$$

Solution is consistent

Span (S) = $(x, y, z) \in R3$

Span (S) = V

So $S = \{V_1, V_2, V_3\}$ is basis on R^3.